Visualization with Seaborn

Python course - Lesson 7

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Previous Lesson Topics

- Connection to SQL
  - Read query data
  - Working with cursor
- Module
- Comparison SQL vs Pandas

Today's Topics

- Introduction to Seaborn package
- Select the correct graph type
- Graph types
- Change style od the graph

Introduction to Seaborn

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Seaborn

- Seaborn is a data visualization library built on top of Matplotlib.
- It is often used because it makes attractive visualizations
- It works easily with Pandas library
- While in Matplotlib you often had to write multiple lines of code to create a plot, Seaborn makes assumptions on what you want which often translates into getting the same plot with one line of code

Select the correct graph type

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Univariate EDA

Univariate

Categorical

Visual EDA:

- sns.countplot ()

Continuous

Visual EDA:

- boxplot ()
- distplot ()
- kdeplot ()

Bivariate EDA

Bivariate

Categorical to Categorical

Visual EDA:

- sns.factorplot ()

Continuous to Continuous

Visual EDA:

- jointplot ()

Continuous to Categorical

Visual EDA:

- Facegrid().map ()

Plots Type

relplot (relational)

Visual EDA:

- sns.scatterplot()
- Sns.lineplot()

distplot (distributions)

Visual EDA:

- histplot ()
- kdeplot ()
- ecfplot ()
- rugplot ()

catplot (categorical)

Visual EDA:

- stripplot ()
- swarmplot ()
- boxplot ()
- violinplot ()
- pointplot ()
- barplot ()

Graph Types

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Creating the Plot

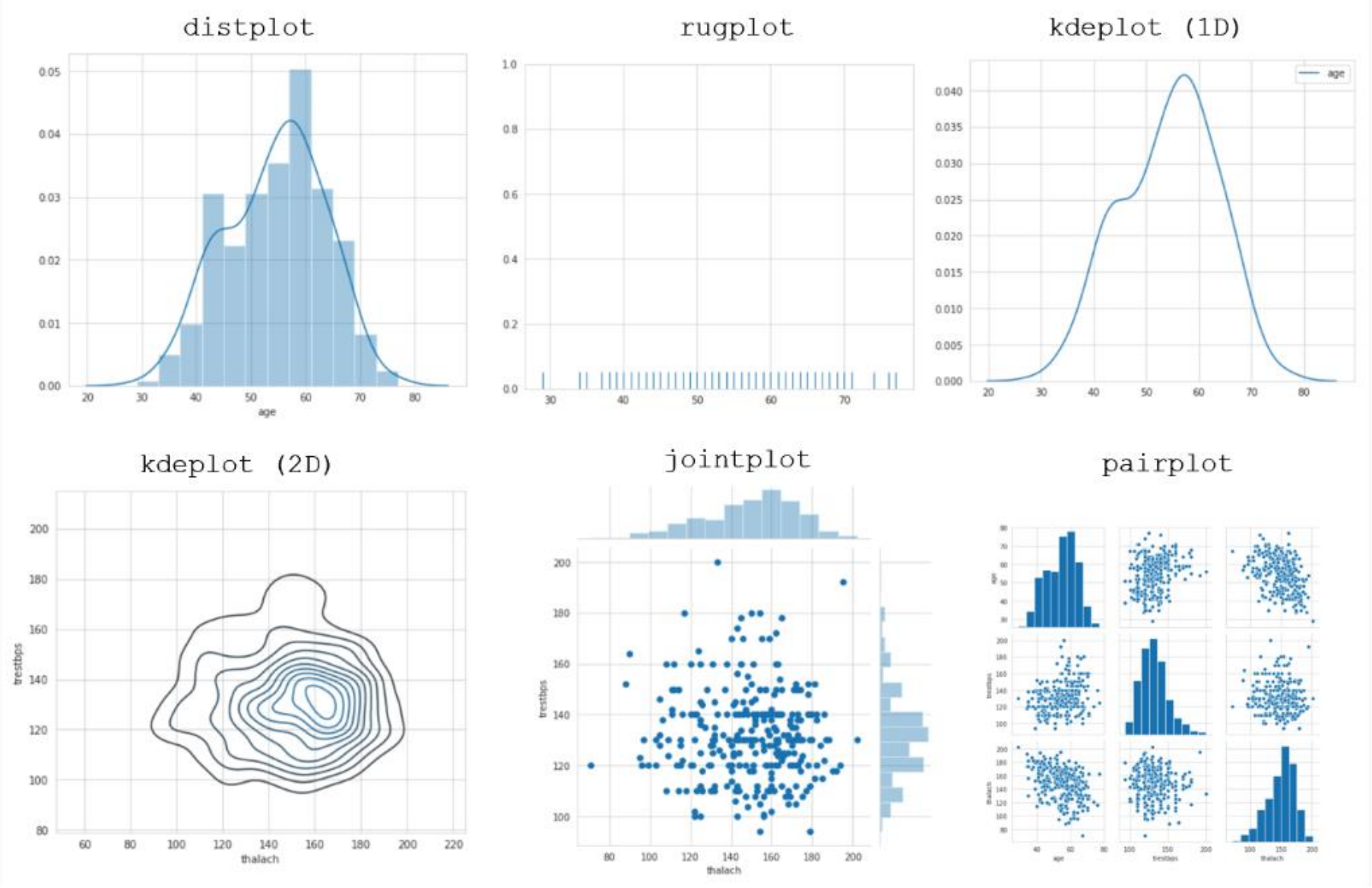
sns.name_of_plot (x, y)

- All plots in seaborn are created with this command, depending on how many dimensions the plot has
- With other command we can change properties of the plot

Graph Types – Distribution Plots

- Distribution plots usually univariate data (data with only one dimension)
- Distribution plots shows where the concentration of data points along a number line is
- Seaborn has adaptations for two-dimensional distribution plots, which shows two distribution plots simultaneously.

Graph Types – Distribution Plots



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Graph Types – Distribution Plots

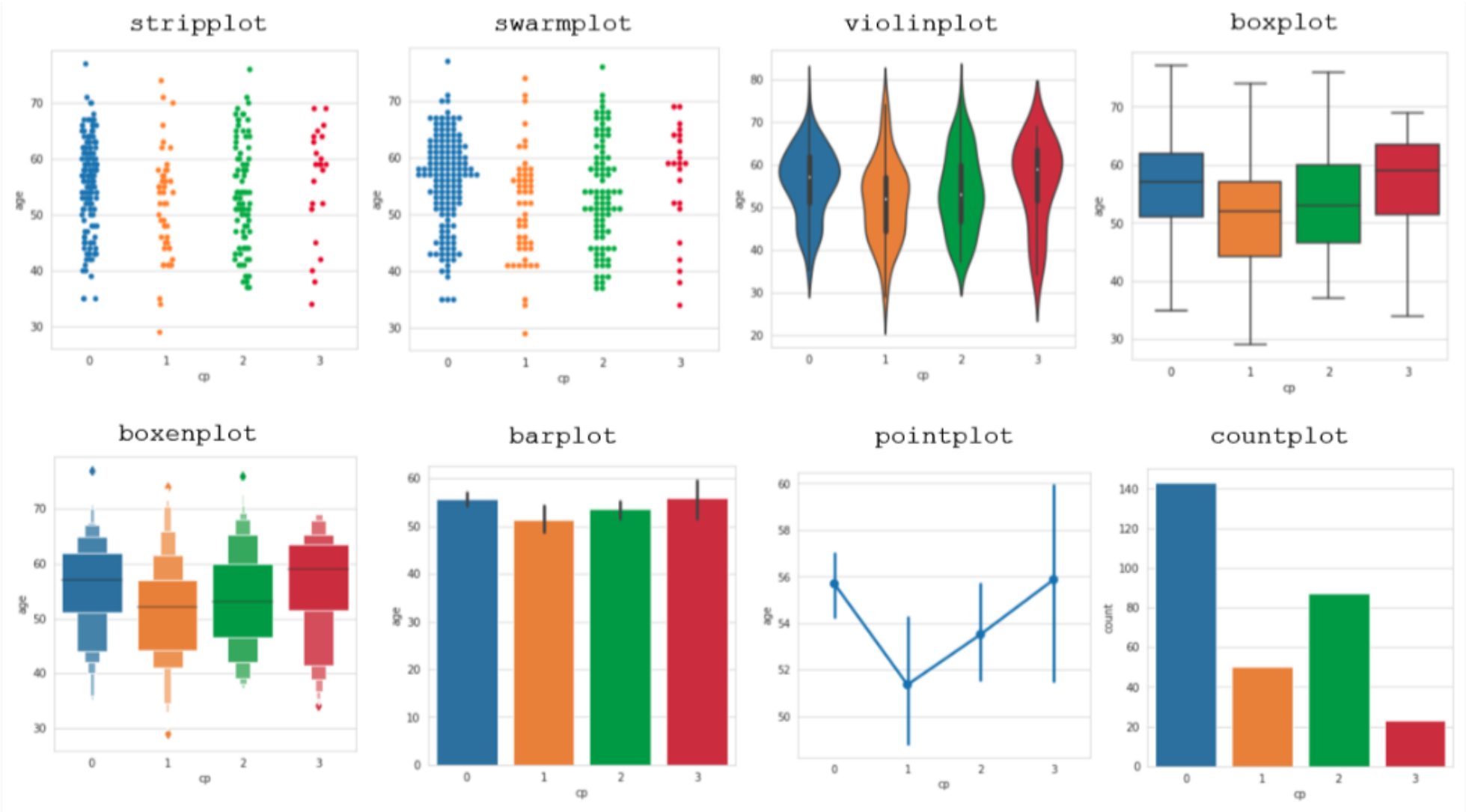
- distplot - a one-dimensional kdeplot with a histogram
- rugplot - ticks in lieu of a data point to show clusters
- kdeplot - when one dimension of data - curve of the distribution.

  When two dimensions of data are given - contour plot
- jointplot - scatterplot with histograms on each side
- Pairplot - plots each dimension of data against each other, and histogram when the variable is plotted against itself

Graph Types – Quantitative and Qualitative

- These plots combine two types of variables – quantitative (continues)and qualitative (categorical)

Graph Types – Quantitative and Qualitative



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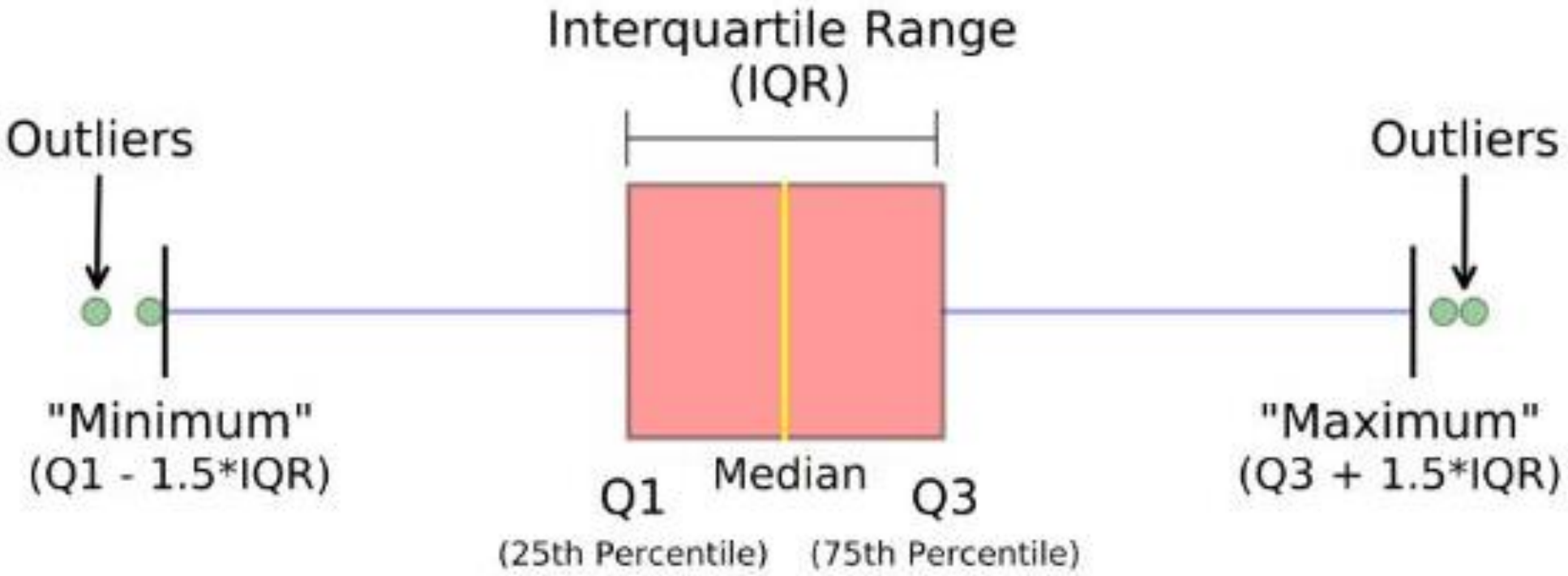
Graph Types – Quantitative and Qualitative

- stripplot - vertical data points horizontally
- Swarmplot - similar to the stripplot, but in a more organized fashion
- violinplot - a distribution on both sides of the quantitative axes
- boxplot - a five-number summary of the data –(minimum, 1st quartile (25th percentile), the median, 3rd quartile (75th percentile), and the maximum
- boxenplot - show tails and a more accurate depiction of the distribution

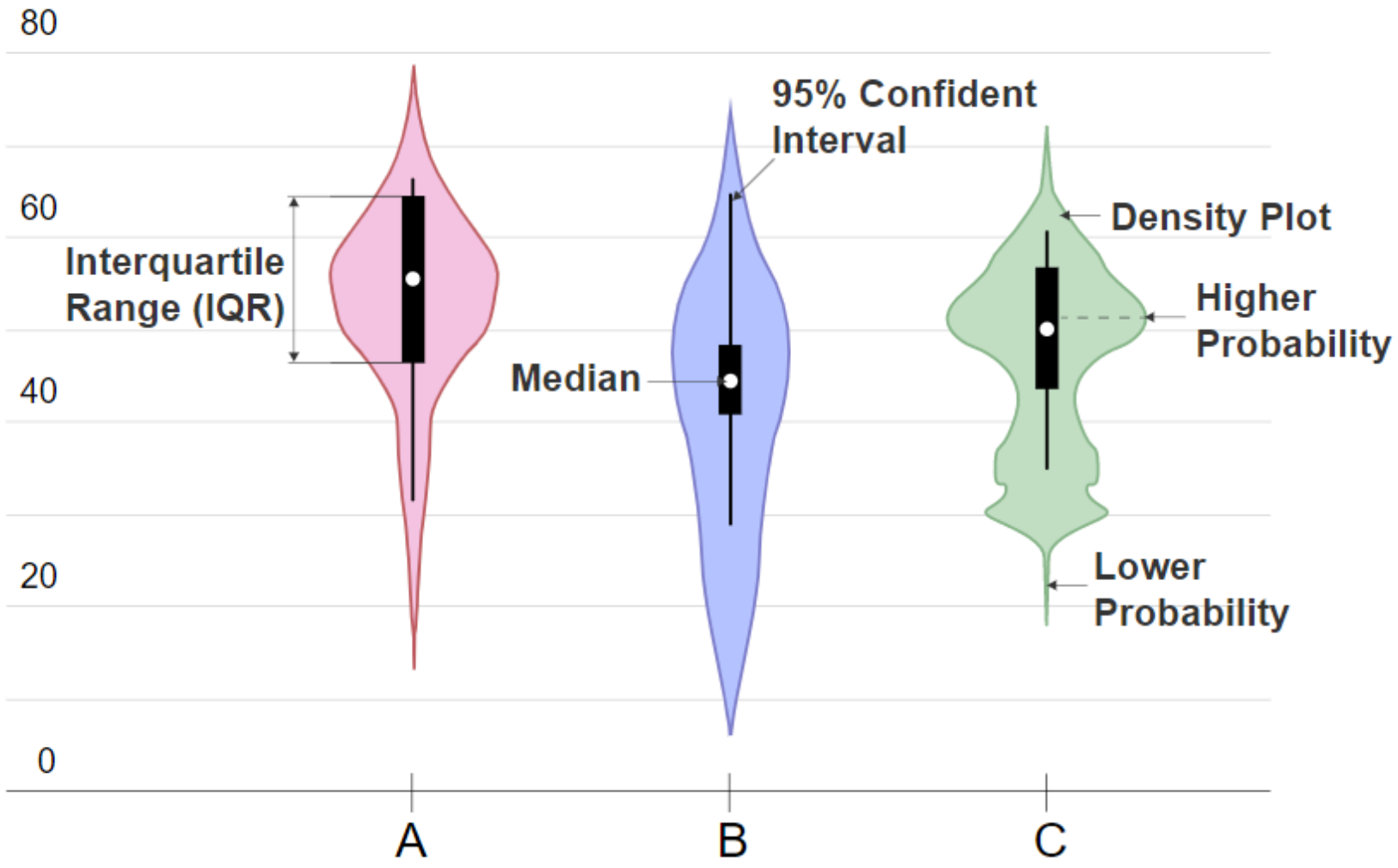
Graph Types – Quantitative and Qualitative

- barplot - displays bars whose height corresponds to the value
- countplot expresses the same visualization as barplot, but takes in only one variable and displays the number of items in each distinct value
- pointplot - tries to find one point (with appropriate error bars) that appropriately represents that array. This plot is great for comparing qualitative variables that are numerical

Boxplot – How to read



Violinplot – How to read



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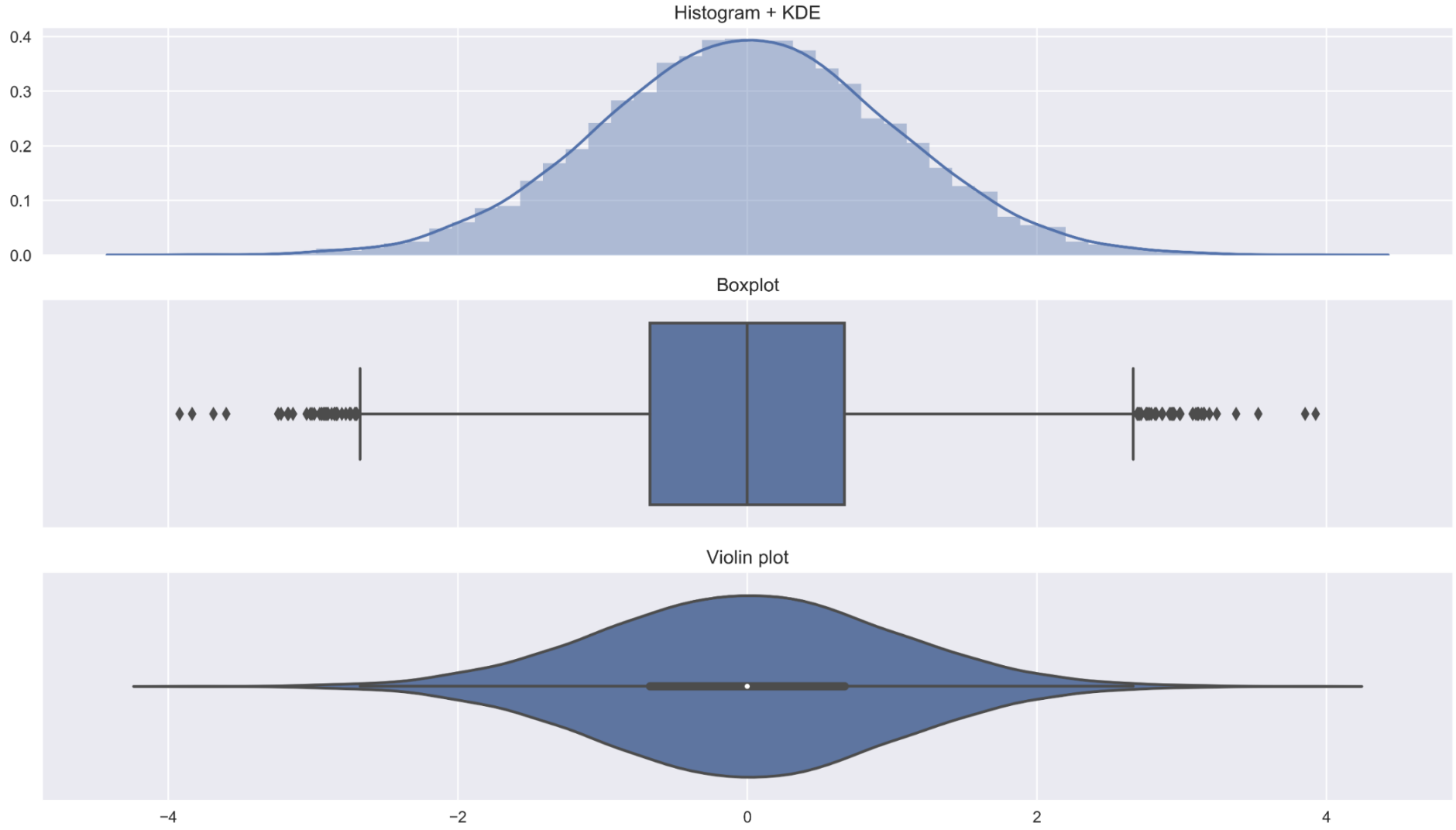
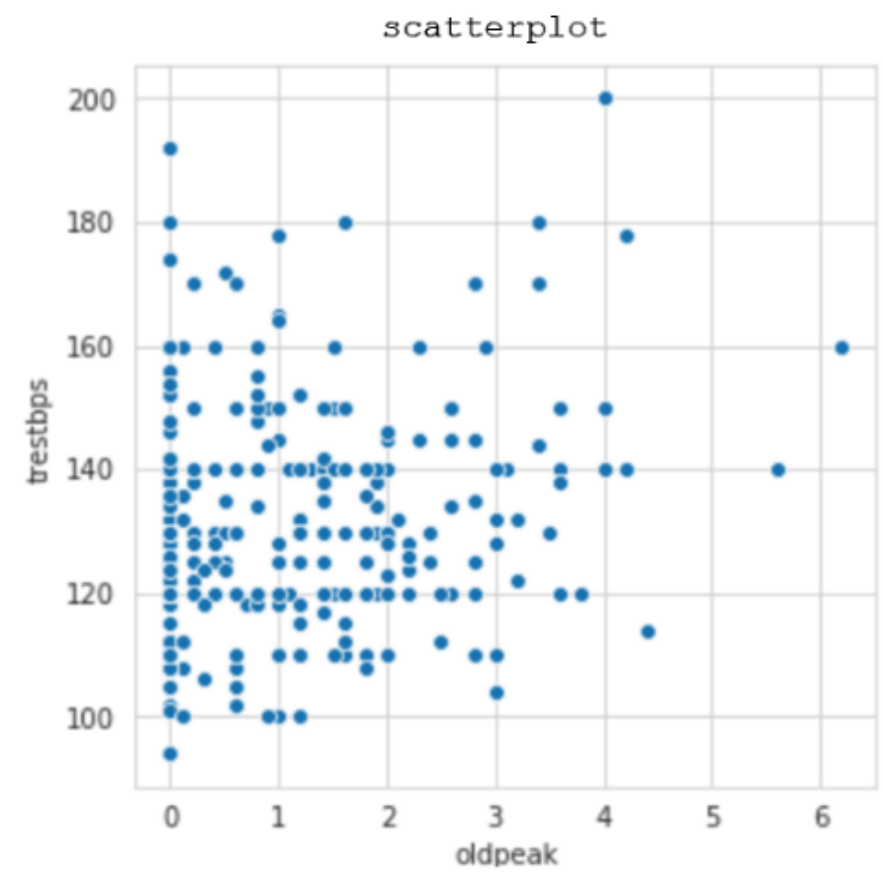
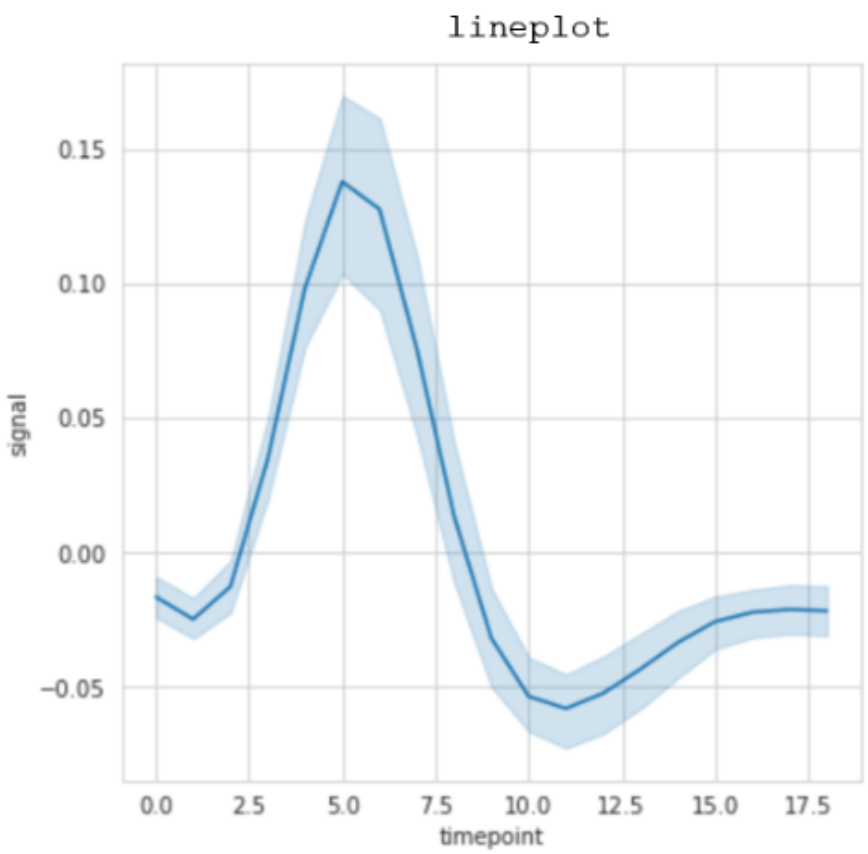
Histogram vs Boxplot vs Violinplot



Graph Types – Quantitative Relationships

These plots show the relationship between two quantitative variables





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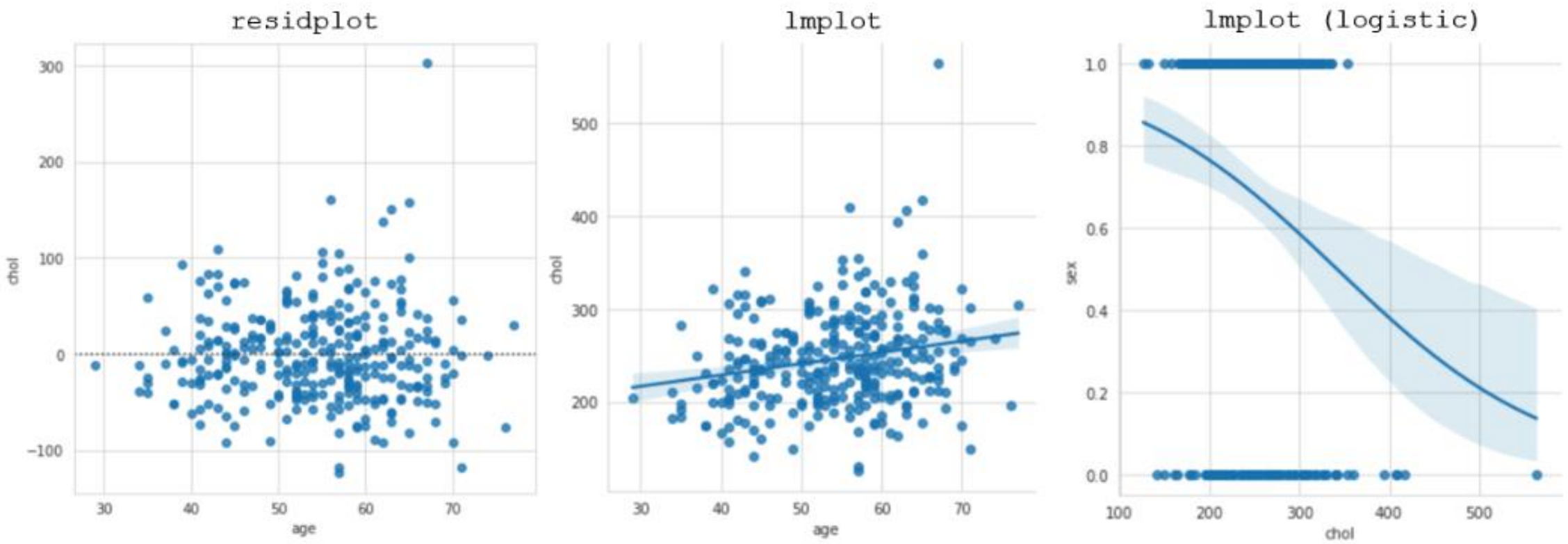
Graph Types – Quantitative relationships

- scatterplot - two quantitative variables against each other
- lineplot - a quantitative variable along a time variable, which may be quantitative or date

Graph Types – Statistical Models

Statistical model visualizations take advantage of statistical models to visualize the nature of the data.



Graph Types – Quantitative and Qualitative

- residplot - displays residuals of a linear regression (how far each data point was off the linear regression fit by Euclidean distance)

- lmplot - displays a linear regression fit with confidence intervals on a scatterplot. This plot has several parameters which can be used to adjust the nature of the plot.

  For instance, setting logistic=True will assume the y-variable is binary and will create a Logistic (sigmoid) regression model

Seaborn Color Palette Codes

Color palette code	Color palette code	Color palette code
summer_r	Set2	deep
Greens_r	Blues	muted
gist_earth_r	BuGn_r	pastel
spring_r	GnBu_d	bright
bwr_r	cubehelix	dark
Reds	BrBG	colorblind
OrRd	RdBu_r	hls
flag	Paired	husl

Matplotlib Markers

Code	Description
'2'	tri_up marker
'3'	tri_left marker
'4'	tri_right marker
's'	square marker
'p'	pentagon marker
'*'	star marker
'h'	hexagon1 marker
'H'	hexagon2 marker
'+'	plus marker
'x'	x marker
'D'	diamond marker
'd'	thin_diamond marker

Code	Description
'-'	solid line style
'--'	dashed line style
'.-'	dash-dot line style
':'	dotted line style
'.'	point marker
','	pixel marker
'o'	circle marker
'v'	triangle_down marker
'^'	triangle_up marker
'>'	triangle_left marker
'<'	triangle_right marker
'1'	tri_down marker

Matplotlib colors

Code	Description
‘b’	blue
‘g’	green
‘r’	red
‘c’	cyan
‘m’	magenta
‘y’	yellow
‘k’	black
‘w’	white

Thank You

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